Specifikáció tétele

Legyen A tetszőleges állapottér.

Legyen S A feletti program, $F \subseteq A \times A$ feladat.

Legyen B az F feladat egy paramétertere

(ekkor $\exists F_1 \subseteq A \times B$ és $\exists F_2 \subseteq B \times A$, hogy

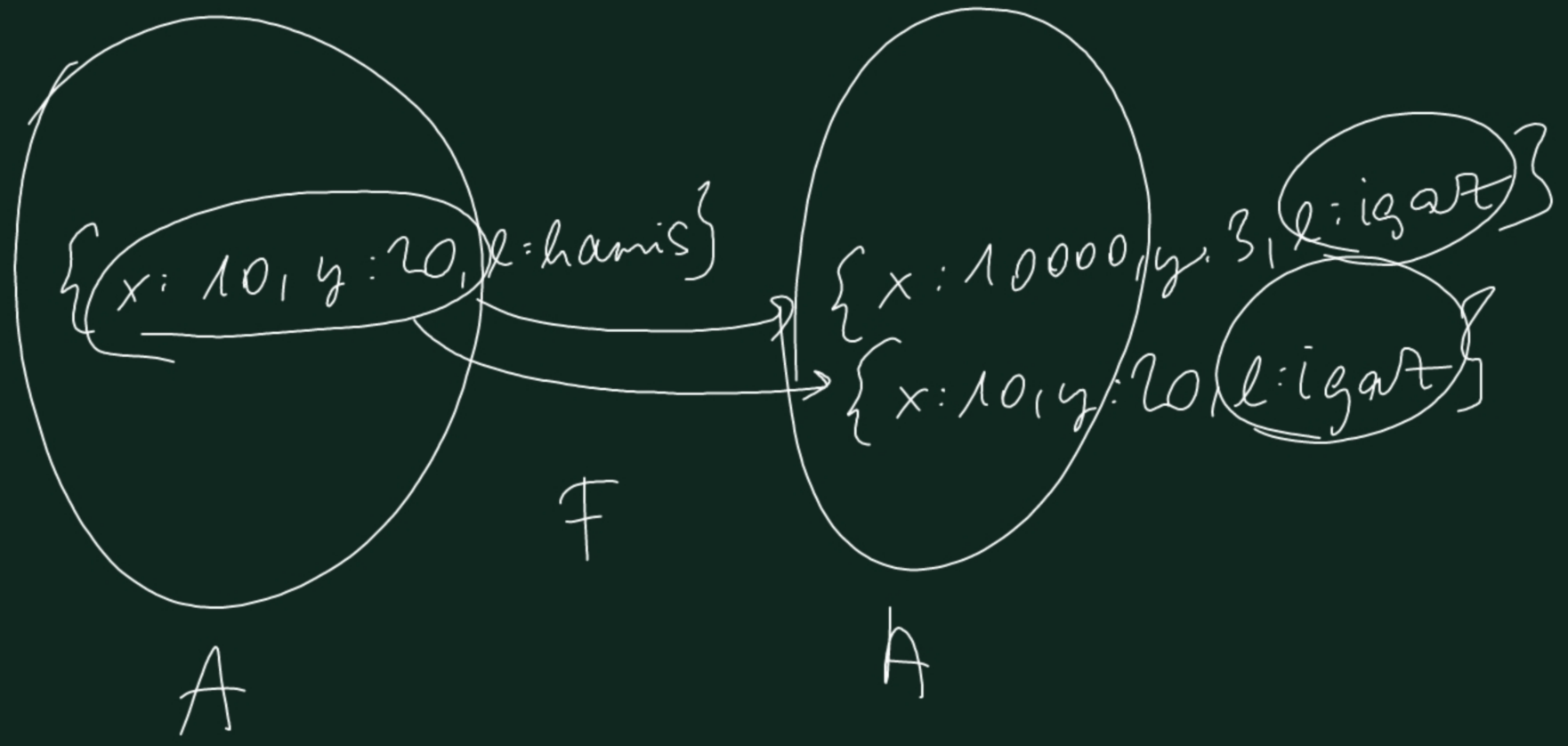
$F = F_2 \circ F_1$). Minden $b \in B$ paraméterhez definiáljuk

a $Q_b : A \to B$, $R_b : A \to B$ logikai függvényeket úgy, hogy

$\ulcorner Q_b \urcorner = F_1^{(-1)}(b)$, $\ulcorner R_b \urcorner = F_2(b)$. Ha $\forall b \in B : Q_b \Rightarrow lf(S, R_b)$,

akkor S megoldja F-et.

$A = (x : \mathbb{N}^+, y : \mathbb{N}^+, l : \mathbb{L})$



Legyen $A = [1..4]$. $S \subseteq A \times (A \cup \{tail\})^{**}$ a következő program:

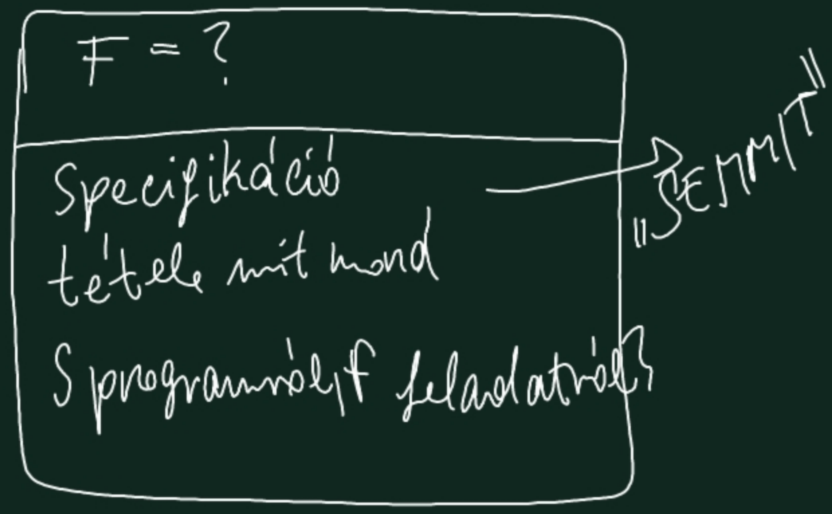
$$S = \left\{ \begin{array}{lll} 1 \longrightarrow \langle 1,2,4,1 \rangle & 1 \rightarrow \langle 1,3,2 \rangle & 2 \rightarrow \langle 2,3 \rangle \\ 3 \rightarrow \langle 3,2 \rangle & 3 \rightarrow \langle 3,4 \rangle & 4 \rightarrow \langle 4,1,3 \rangle \end{array} \right\}$$

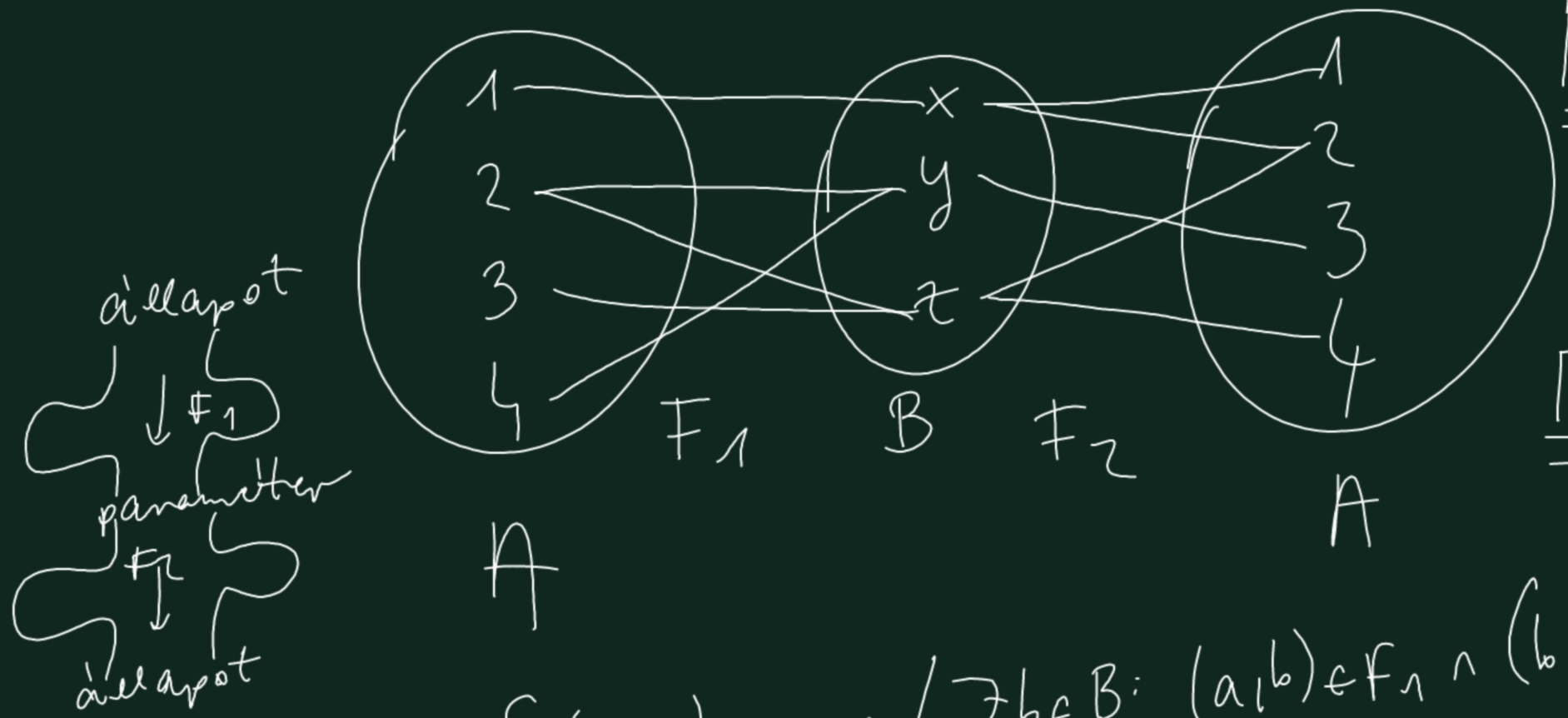
Legyen $B = \{x, y, z\}$ az $F \subseteq A \times A$ feladat egy paramétertere.

Adott továbbá:

$$F_1 = \{(1,x),(2,y),(2,z),(3,z),(4,y)\}$$

$$F_2 = \{(x,1),(x,2),(y,3),(z,2),(z,4)\}.$$





$[Q_x] = \{1\}$

$[Q_y] = \{2, 4\}$

$[Q_z] = \{2, 3\}$

$$F = F_2 \circ F_1 = \{(a, c) \in A \times A \,/\, \exists b \in B : (a, b) \in F_1 \wedge (b, c) \in F_2\}$$

$$F = \{(1,1), (1,2), (2,3), (2,2), (2,4), (3,2), (3,4), (4,3)\}$$

$$\lceil R_x \rceil = \{1,2\}$$

$$\lceil R_y \rceil = \{3\}$$

$$\lceil R_z \rceil = \{2,4\}$$

$$a_x \Longrightarrow \mathcal{U}(S, R_x)$$

$$\lceil a_x \rceil \subseteq \lceil \mathcal{U}(S, R_x) \rceil$$

$$\{1\} \subseteq \{a \in A \mid a \in D_{p(S)} \wedge p(S)(a) \subseteq \lceil R_x \rceil\}$$

$$\{1\} \subseteq \{1\} \checkmark$$

$$a_y \Rightarrow \mathcal{U}(S, R_y)$$

$$\{2,4\} \subseteq \{a \in A \mid a \in D_{p(S)} \wedge p(S)(a) \subseteq \lceil R_y \rceil\}$$

$$\{2,4\} \subseteq \{2,4\} \checkmark$$

$$a_z \Rightarrow \mathcal{U}(S, R_z)$$

$$\{2,3\} \not\subseteq \{3\}$$

$A = (x : \mathbb{R},\ y : \mathbb{R},\ z : \mathbb{R})$

$B = (x' : \mathbb{R},\ y' : \mathbb{R})$

$Q = (x = x' \wedge y = y' \wedge x' > 5)$

$R = (Q \wedge x > y \rightarrow z = x)$

Relacionat pojem specifikace

$Q_{\{x' : 4,\ y' : 2\}} = \emptyset$

$\in B$

$F_1(\{x : 8,\ y : 5,\ z : 7\}) = \{\{x' : 8,\ y' : 5\}\}$

$F_2(\{x' : 8,\ y' : 5\}) = \{\{x : 8,\ y : 5,\ z : 8\}\}$

$F(\{x : 4,\ y : 2,\ z : 1\}) = \emptyset$

$F(\{x : 8,\ y : 5,\ z : 7\}) = \{\{x : 8,\ y : 5,\ z : 8\}\}$

$F(\{x : 9,\ y : 3,\ z : 10\}) = \{\{x : 9,\ y : 3,\ z : 9\}\}$

$F(\{x : 10,\ y : 20,\ z : 5\}) = \{\{x : 10,\ y : 20,\ z : \varphi\} \mid \varphi \in \mathbb{Z}\}$

$A = (x : \mathbb{N}, d : \mathbb{N})$

$B = (x' : \mathbb{N})$

$Q = (x = x' \wedge x > 0)$

$R = (Q \wedge 10^{d-1} \leq x < 10^{d})$

$\boxed{Q_{\{x' : 6854\}}} = \{\{x : 6854, d : \delta\} \mid \delta \in \mathbb{N}\}$

$\boxed{R_{\{x' : 6854\}}} = \{\{x : 6854, d : 4\}\}$

$F(\{x : 6854, d : 2\}) = \{\{x : 6854, d : 4\}\}$

$F = \{(a, b) \in A \times A \mid x(a) > 0 \wedge x(b) = x(a) \wedge 10^{d(b)-1} \leq x(a) < 10^{d(b)}\}$

$F(\{x : 7267363, d : 2\}) = \{\{x : 7267363, d : 7\}\}$

$$A = (x:\mathbb{R},\ y:\mathbb{R},\ z:\mathbb{R})$$

$$B = (x':\mathbb{R},\ y':\mathbb{R})$$

$$Q = (x = x' \wedge y = y')$$

$$R = ((z = x' \vee z = y') \wedge z \geq x' \wedge z \geq y')$$

$$F(\{x:6,\ y:5,\ z:3\}) = \{\{x:\alpha,\ y:\beta,\ z:6\} \mid \alpha \in \mathbb{R},\ \beta \in \mathbb{R}\}$$

$$F_1(\{x:6,\ y:5,\ z:3\}) = \{\{x':6,\ y':5\}\}$$

$$F_2(\{x':6,\ y':5\}) = \{\{x:\alpha,\ y:\beta,\ z:6\} \mid \alpha \in \mathbb{R},\ \beta \in \mathbb{R}\}$$

$$[Q_{\{x':6,\ y':5\}}] = \{\{x:6,\ y:5,\ z:\alpha\} \mid \alpha \in \mathbb{R}\}$$

$$[R_{\{x':6,\ y':5\}}] = \{\{x:\alpha,\ y:\beta,\ z:6\} \mid \alpha \in \mathbb{R},\ \beta \in \mathbb{R}\}$$

$$A = (x : \mathbb{N}, y : \mathbb{N}, z : \mathbb{N})$$

$$B = (x' : \mathbb{N}, y' : \mathbb{N})$$

$$Q = (x = x' \wedge y = y' \wedge x \leq y + 1)$$

$$R = (Q \wedge \mathrm{prim}(z) \wedge \min(|x - z|, |y - z|) \leq 3)$$

7 8 9 10 11 12 13 16 17 18 19 20 21 22

$$F(\{x : 10, y : 19, z : 13\}) = \{\{x : 10, y : 19, z : \varphi\} \mid \varphi \in \{7, 11, 13, 17, 19\}\}$$

$$F_1(\{x : 10, y : 19, z : 13\}) = \{\{x' : 10, y' : 19\}\}$$

$$F_2(\{x' : 10, y' : 19\}) = \{\{x : 10, y : 19, z : \varphi\} \mid \varphi \in \{7, 11, 13, 17, 19\}\}$$

$A = (n : \mathbb{N}, p : \mathbb{N})$

$B = (n' : \mathbb{N})$

$Q = (n = n')$

$R = (Q \wedge prim(p) \wedge \forall i \in \mathbb{N}^+ : prim(i) \rightarrow |n - i| \geq |n - p|)$

$F(\{n:10, p:1\}) = \{\{n:10, p:11\}\}$

$\{n':10\}$

$F(\{n:9, p:5\}) = \{\{n:9, p:7\}, \{n:9, p:11\}\}$

$\{n':9\}$

$$F(\{x:26, y:34, z:31\})$$

$$F_1(\{x:26, y:34, z:31\}) = \{\{x':26, y':34\}\}$$

$$F_2(\{x':26, y':34\}) = \{\{x:26, y:34, z:4\} \mid 4 \in \{23, 29, 31, 37\}\}$$

23	31
24	32
25	33
26	34
27	35
28	36
29	37